

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) netflix_titles.csv
netflix_titles.csv(text/csv) - 3399671 bytes, last modified: 6/6/2026 - 100% done
Saving netflix_titles.csv to netflix_titles (1).csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
```

```
df = pd.read_csv('netflix_titles.csv')
```

```
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmmaker Kirsten Johnson...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thabane...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	A gripping story of a woman who crosses paths with a powerful...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, ...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International	To protect family from...

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

(8807, 12)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
df.isnull().sum()
```

```

0
show_id    0
type       0
title      0
director   2634
cast       825
country    831
date_added 10
release_year 0
rating     4
duration   3
listed_in  0
description 0

```

dtype: int64

```

df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Unknown')
df['rating'] = df['rating'].fillna(df['rating'].mode()[0])

```

df.dropna(inplace=True)

df.drop_duplicates(inplace=True)

df.shape

(8794, 12)

df.describe(include='all')

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
count	8794	8794	8794	8794	8794	8794	8794	8794.000000	8794	8794	8794	8794
unique	8794	2	8794	4528	7682	749	1766	NaN	14	220	513	8794
top	s8807	Movie	Zubaan	Unknown	Unknown	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity, It abandoned prop
freq	1	6128	1	2624	825	2809	109	NaN	3209	1793	362	
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.183534	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.823527	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN	NaN

df['type'].value_counts()

```

count
type
Movie    6128
TV Show  2666

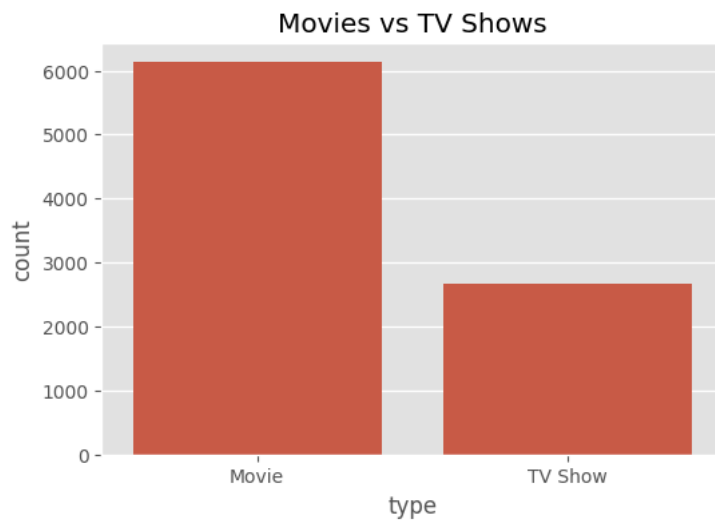
```

dtype: int64

```

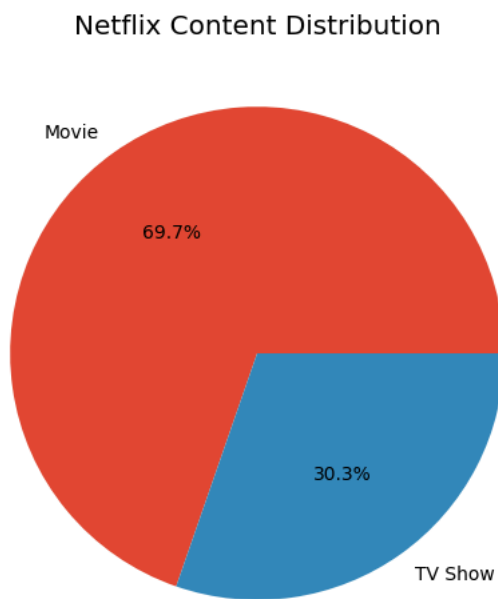
plt.figure(figsize=(6,4))
sns.countplot(data=df, x='type')
plt.title('Movies vs TV Shows')
plt.show()

```



```
df['type'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(6,6)
)

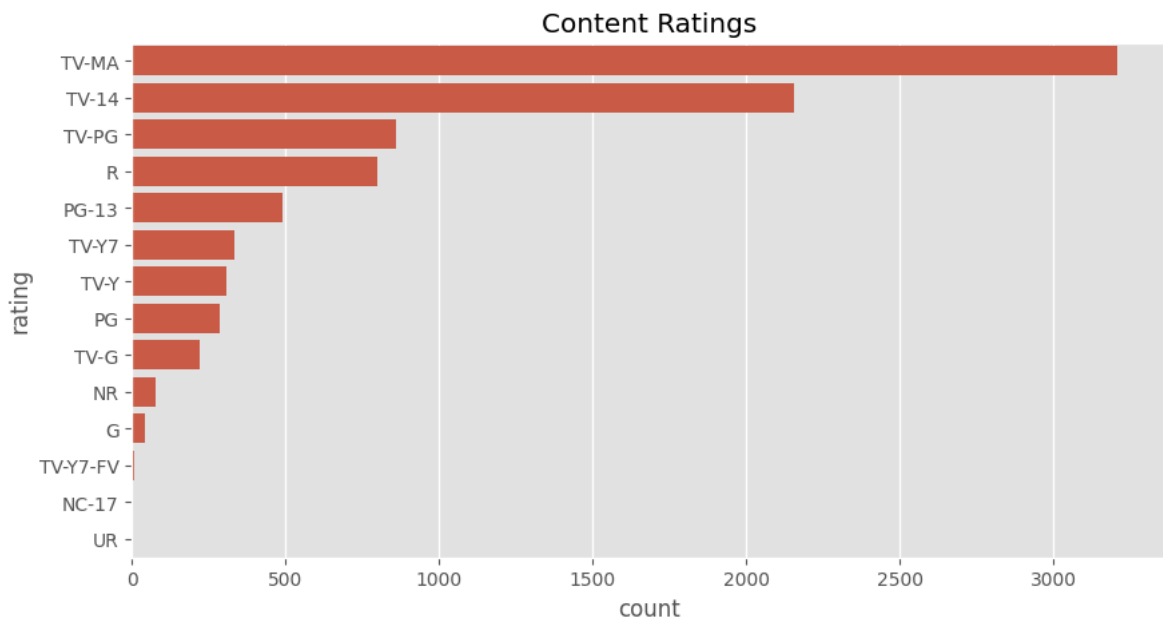
plt.ylabel('')
plt.title('Netflix Content Distribution')
plt.show()
```



```
plt.figure(figsize=(10,5))

sns.countplot(
    y='rating',
    data=df,
    order=df['rating'].value_counts().index
)

plt.title('Content Ratings')
plt.show()
```



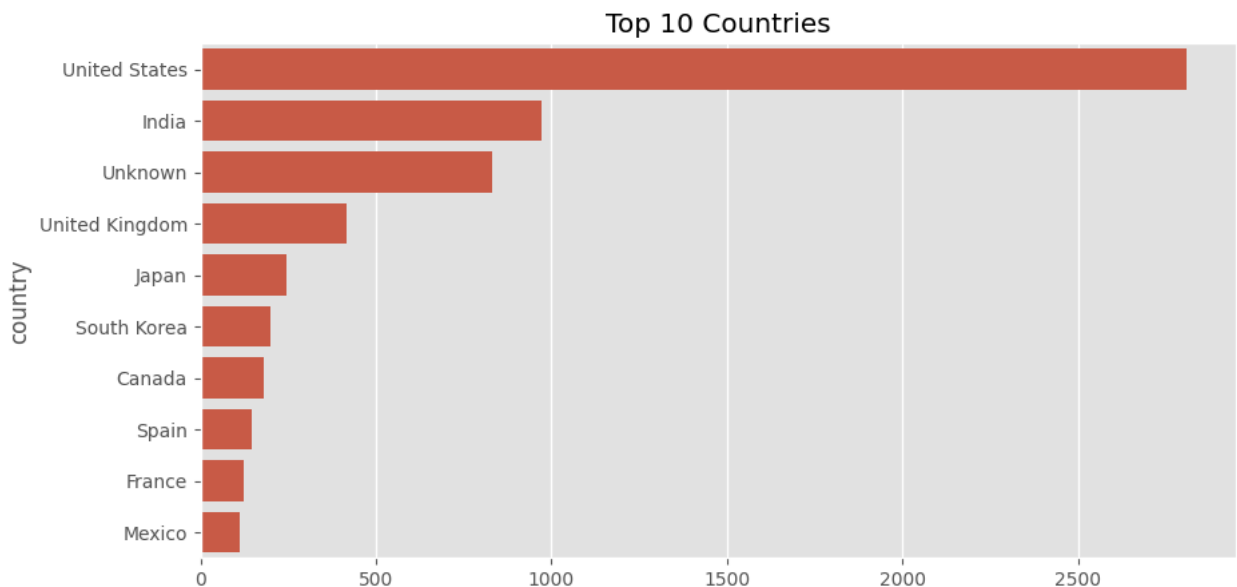
```
top_countries = df['country'].value_counts().head(10)
```

```
plt.figure(figsize=(10,5))
```

```
sns.barplot(
    x=top_countries.values,
    y=top_countries.index
)
```

```
plt.title('Top 10 Countries')
```

```
plt.show()
```

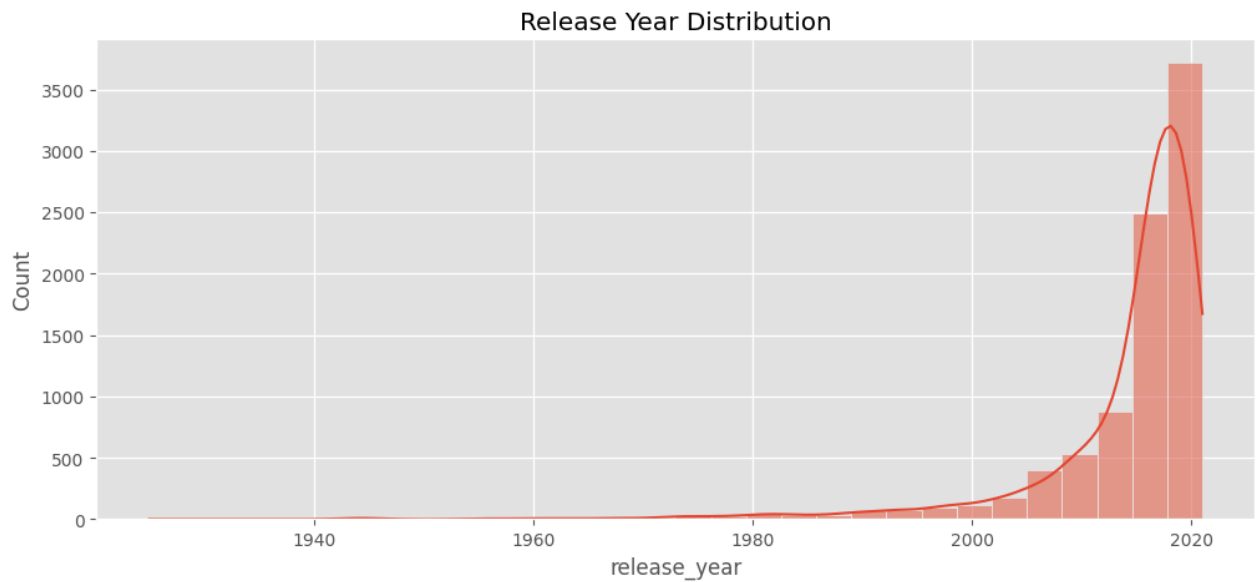


```
plt.figure(figsize=(12,5))
```

```
sns.histplot(
    df['release_year'],
    bins=30,
    kde=True
)
```

```
plt.title('Release Year Distribution')
```

```
plt.show()
```



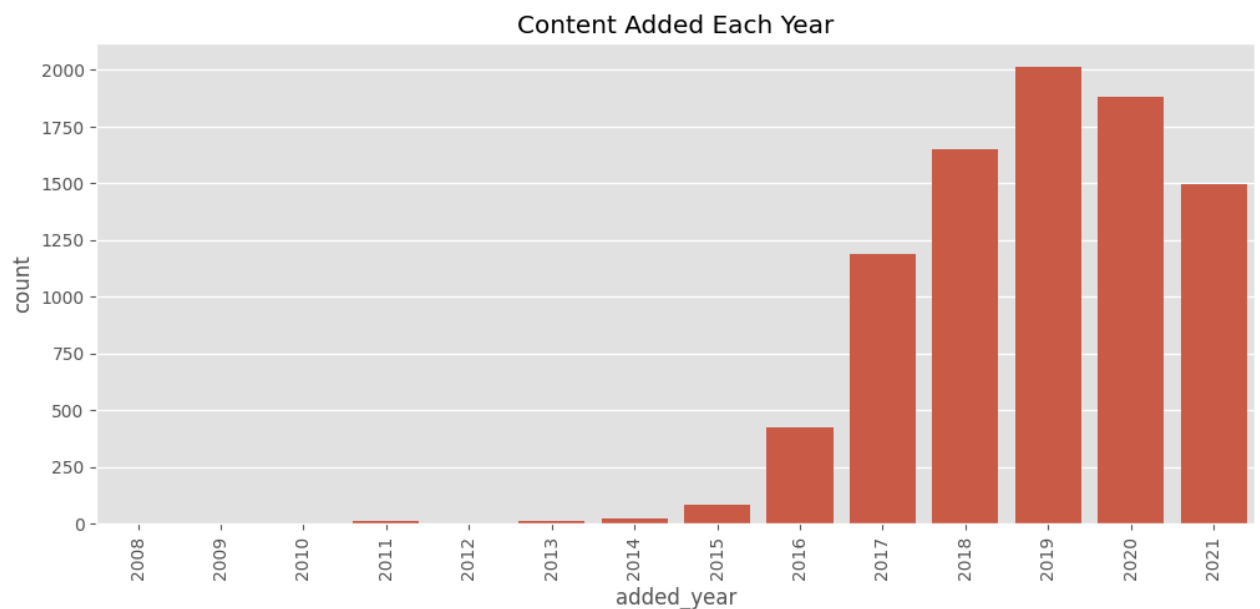
```
df['date_added'] = pd.to_datetime(df['date_added'], format='mixed')
```

```
df['added_year'] = df['date_added'].dt.year
```

```
plt.figure(figsize=(12,5))

sns.countplot(
    x='added_year',
    data=df,
    order=sorted(df['added_year'].dropna().unique())
)

plt.xticks(rotation=90)
plt.title('Content Added Each Year')
plt.show()
```



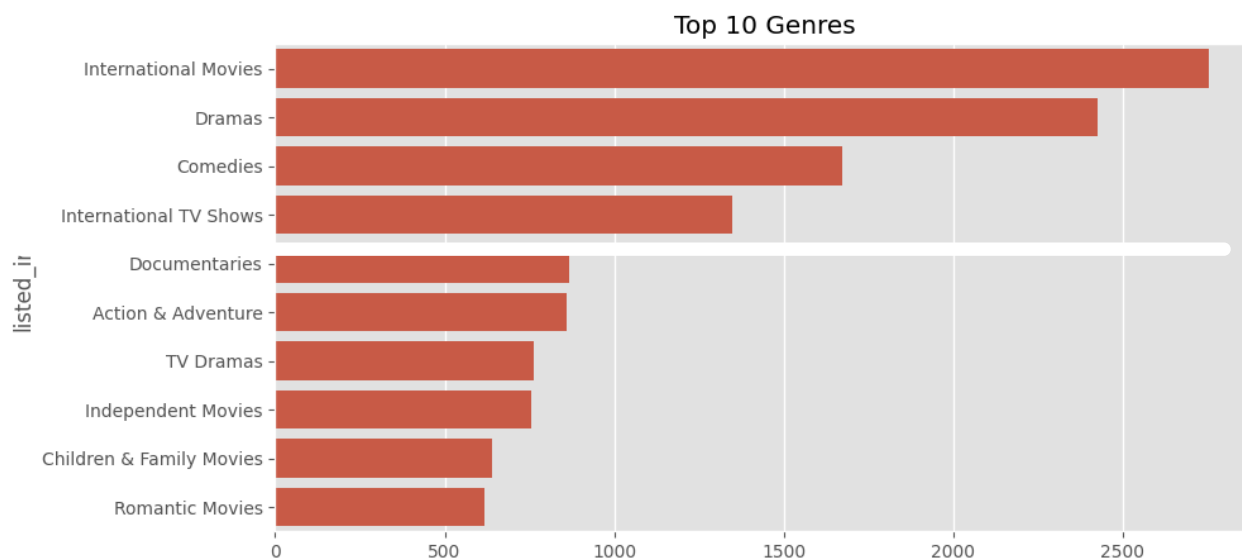
```
genres = df['listed_in'].str.split(', ').explode()

top_genres = genres.value_counts().head(10)

plt.figure(figsize=(10,5))

sns.barplot(
    x=top_genres.values,
    y=top_genres.index
)
```

```
plt.title('Top 10 Genres')
plt.show()
```



```
numeric_df = df[['release_year']]
```

```
corr = numeric_df.corr()
```

```
corr
```

	release_year
release_year	1.0

```
plt.figure(figsize=(5,4))
```

```
sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm'
)
```

```
plt.title('Correlation Matrix')
plt.show()
```

